Upgrading the Sparkfun Twist to version 1.2

Programming the Sparkfun Twist RGB Rotary Encoder from its source code using an Arduino Uno as ISP Programmer

Note that October 2018 is the date of the current (June 2026) source code Qwiic_Twist.ino

1. Install Arduino IDE 1.8.19 – remember it is 2018 not 2026
2. Install the library PinChangeInterrupt from NicoHood 1.2.9
3. Install the board SpenceKonde ATTinyCore 1.3.3 – the more recent versions do not compile (for me).
4. *Optionally install the* Sparkfun Qwiic Twist Arduino library
5. Copy the folder Qwiic_Twist/Firmware/Qwiic_Twist to Documents/Arduino containing 3 files Qwiic_Twist.ino and nvm.h (and interrupts.ino).
6. Open the installed library PinChangeInterrupt and edit the very last section of PinChangeInterruptSettings.h. Add a comment shown below in red. If this is not changed the encoder will not work on the Twist.

```
#if defined(__AVR_ATtiny24__) || defined(__AVR_ATtiny44__) ||
defined(__AVR_ATtiny84__)
// Port1 is connected to reset, crystal and Pin Interrupt 0
// Deactivate it by default
#if defined(PCINT_ENABLE_PORT1)
// #undef PCINT_ENABLE_PORT1
#endif
#endif
```

7. Plug in the Arduino Uno in the PC USB port, select the Uno board, select its COM port, and select example 11 ArduinoISP, and upload it to the Uno. Note the comments in the example about which Uno pins to use for a connection to an ISP header.

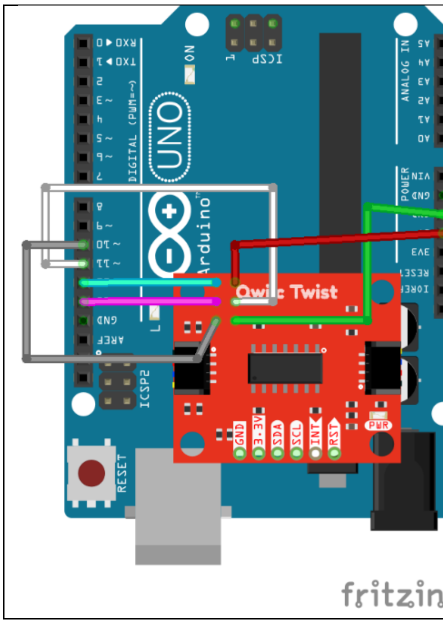
```
// This sketch turns the Arduino into a AVRISP using the following Arduino pins:
//
// Pin 10 is used to reset the target microcontroller.
//
// By default, the hardware SPI pins MISO, MOSI and SCK are used to communicate
// with the target. On all Arduinos, these pins can be found
// on the ICSP/SPI header:
//
//               MISO °. . 5V (!) Avoid this pin on Due, Zero...
//               SCK   . . MOSI
//                     . . GND
//
// On some Arduinos (Uno,...), pins MOSI, MISO and SCK are the same pins as
// digital pin 11, 12 and 13, respectively. That is why many tutorials instruct
// you to hook up the target to these pins. If you find this wiring more
// practical, have a define USE_OLD_STYLE_WIRING. This will work even when not
// using an Uno. (On an Uno this is not needed).
```

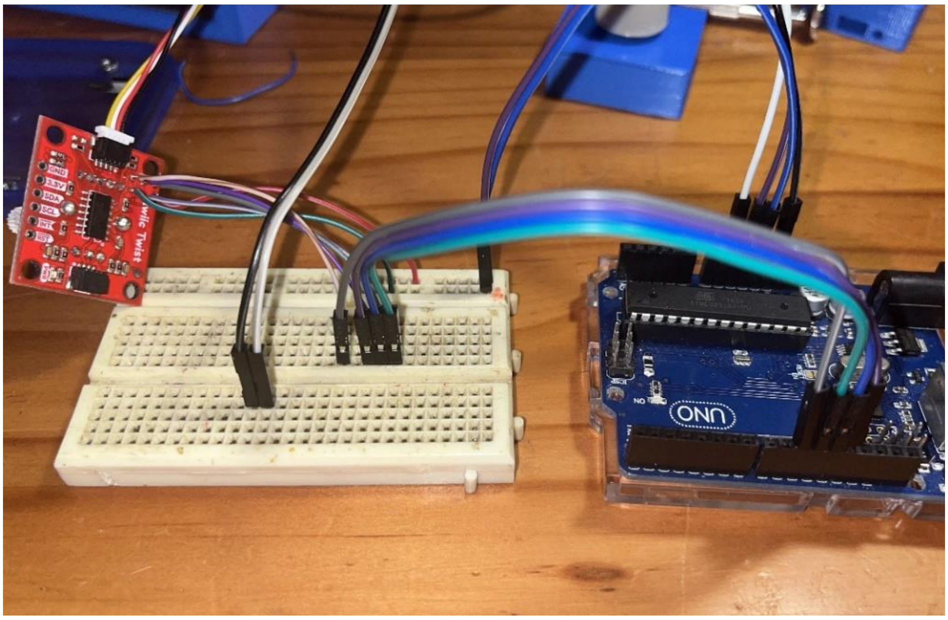
8. Solder six wires (but see below for alternatives), to the ISP header pads on the Sparkfun Twist, and connect them to the Arduino Uno as below – although the Qwiic connector is shown in the photo it is unconnected on the other side.

Do not power the Sparkfun Twist with 5v from the Uno AND also have the Twist connected to the Pico macropad!

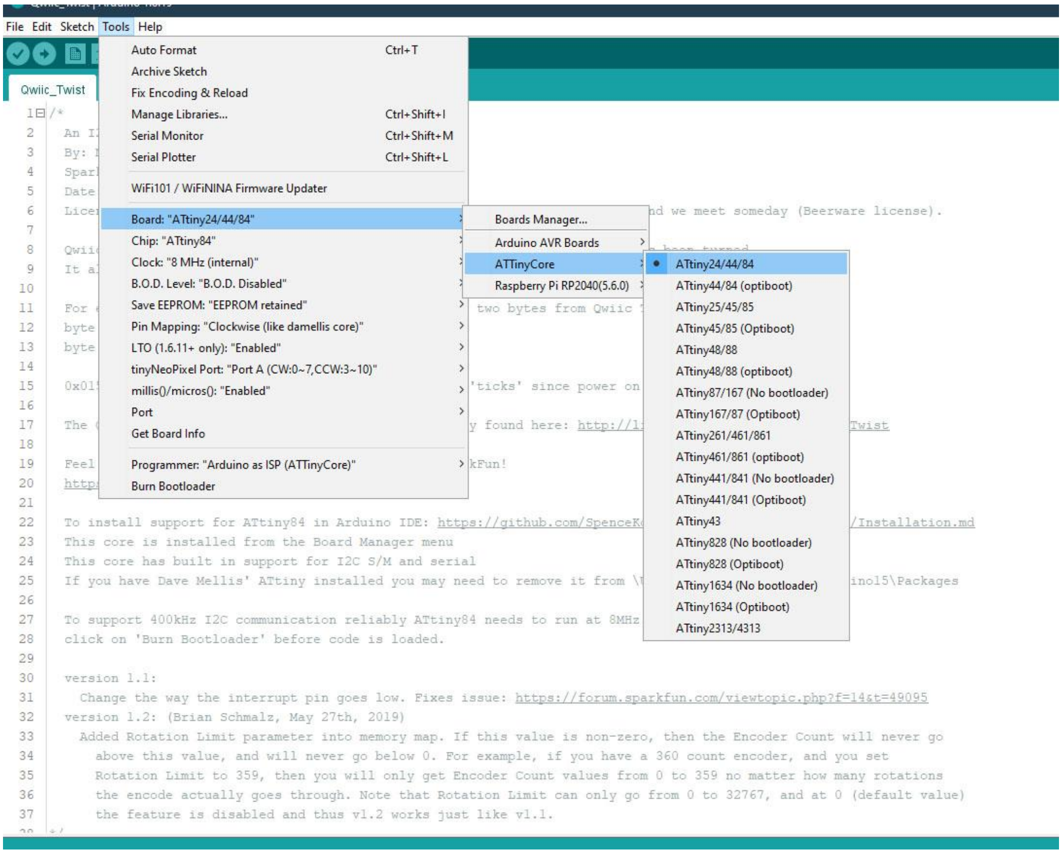
Note that you can get away with only soldering one wire (BLUE or MISO), on the tiny ISP header by using the (bigger) 3v3 GND RST SDA and SCL connectors. If the Qwiic connector is used you only need to solder RST and MISO – study the schematic on the Sparkfun product page.

Uno	Twist
10	RST (Reset)
11	SDA (MOSI)
12	BLUE Twist LED (MISO) (ISP header pin 1 – top left when viewed from PCB underside)
13	SCL (SCLK)
5v	3v3
GND	GND

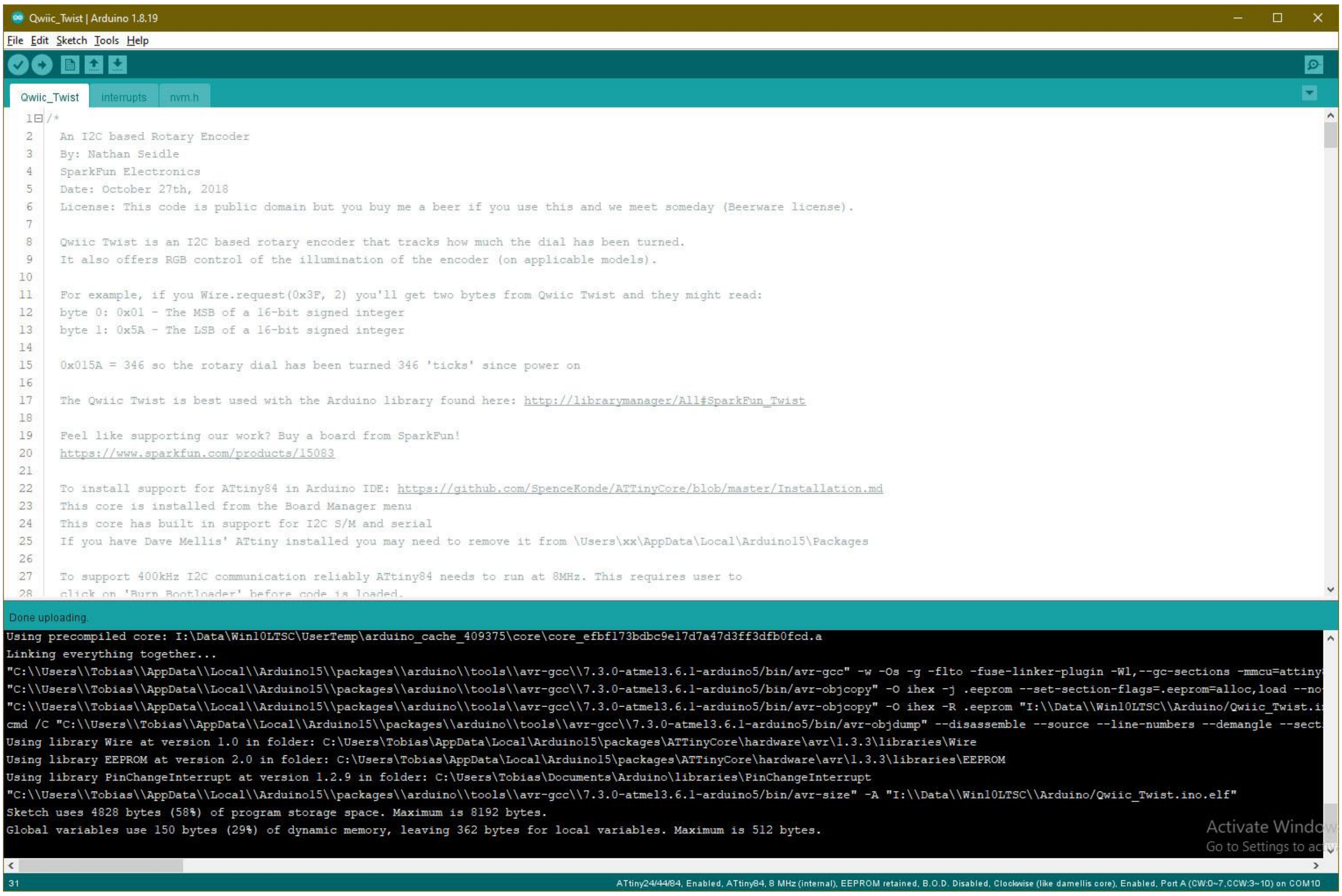




9. Connect the Uno to the PC USB port, double-click on the file Qwiic_Twist.ino and when the Arduino IDE is open select Tools – Board – AttinyCore, and then select ATtiny24/44/84 as below:



10. Make sure the Arduino Uno is **listed and selected** as the ISP programming device under Tools – Programmer and check if the Uno COM port is still selected. Then click on the compile icon, and then click the upload icon - the Twist encoder shaft will turn purple or blue for about 20 seconds and then may also turn white when the upload is completed. The Arduino IDE should look similar to below:



11. The build files are in the location shown in the File – Preferences and are included here in the Build.zip folder – this includes the new Qwiic_Twist.ino.hex file that were uploaded to the ATTiny84 on the Twist. Use a suitable tool to check the Twist version which is now 1.2.